

CLINICALLY-ANCHORED MENTAL HEALTH VITAL SIGNS MONITOR WITH AUTOENCODER ANOMALY DETECTION

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ABSTRACT

A system for generating and monitoring composite mental health vital signs. The system utilizes Principal Component Analysis (PCA) to distill high-dimensional assessment data, journal entries, and medication logs into discrete, clinically-anchored indices: Anxiety Burden Score (ABS), Sleep Quality Index (SQI), Mood Stability Index (MSI), and Cognitive Function Score (CFS). An autoencoder neural network continuously monitors the multidimensional trajectory of these vital signs, detecting subtle, non-linear anomalies that deviate from the patient's established baseline.

BACKGROUND OF THE INVENTION

Physical medicine relies on standardized vital signs (blood pressure, heart rate) to quickly assess patient status. Psychiatry lacks equivalent objective, continuous metrics. Clinicians must manually synthesize diverse data points (questionnaires, patient narratives, sleep logs) to gauge progress, which is time-consuming and prone to subjective bias.

SUMMARY OF THE INVENTION

The present invention establishes a standardized 'Vital Signs' framework for mental health. By applying PCA to diverse data streams, the system generates continuous, 0-100 scale indices for core psychiatric dimensions. To detect complex deteriorations, an autoencoder neural network learns the patient's unique multi-dimensional baseline; when the reconstruction error of the current vital sign vector exceeds a threshold, an anomaly alert is generated.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Vital Signs Architecture, showing the PCA Engine (110) and Autoencoder (120). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core autoencoder anomaly detection algorithm (310). FIG. 4 displays performance metrics. FIG. 5 compares the present invention against manual clinical synthesis. FIG. 6 details the safety verification pipeline. FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A system for monitoring mental health vital signs, comprising: a processor configured to: receive heterogeneous psychiatric data streams; execute a Principal Component Analysis (PCA) algorithm to reduce the dimensionality of the data streams into a plurality of composite indices, including at least a mood stability index and an anxiety burden score; scale the composite indices to a standardized numerical range anchored to normative clinical populations; execute an autoencoder neural network comprising a Long Short-Term Memory (LSTM) architecture trained on a historical baseline of the patient's composite indices; calculate a reconstruction error for a current vector of the composite indices; and generate an anomaly alert when the reconstruction error exceeds a dynamic threshold.

FIG. 1 - Vital Signs Monitor System Architecture

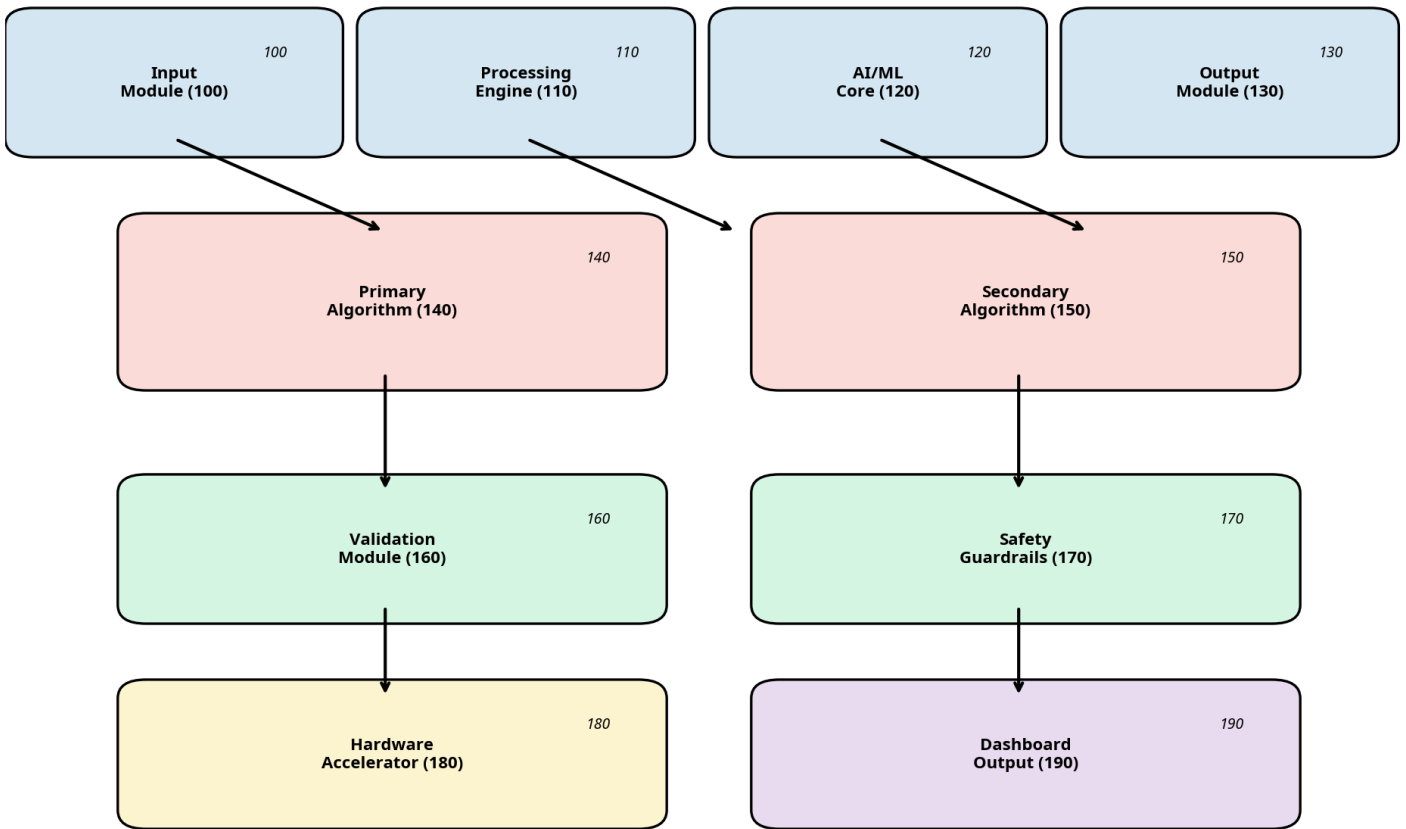


FIG. 2 - Vital Signs Monitor Data Flow Pipeline

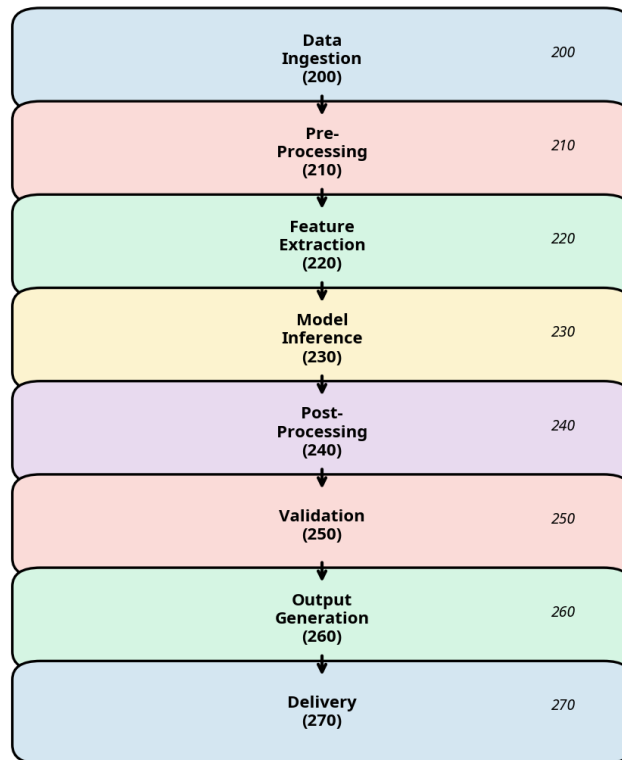
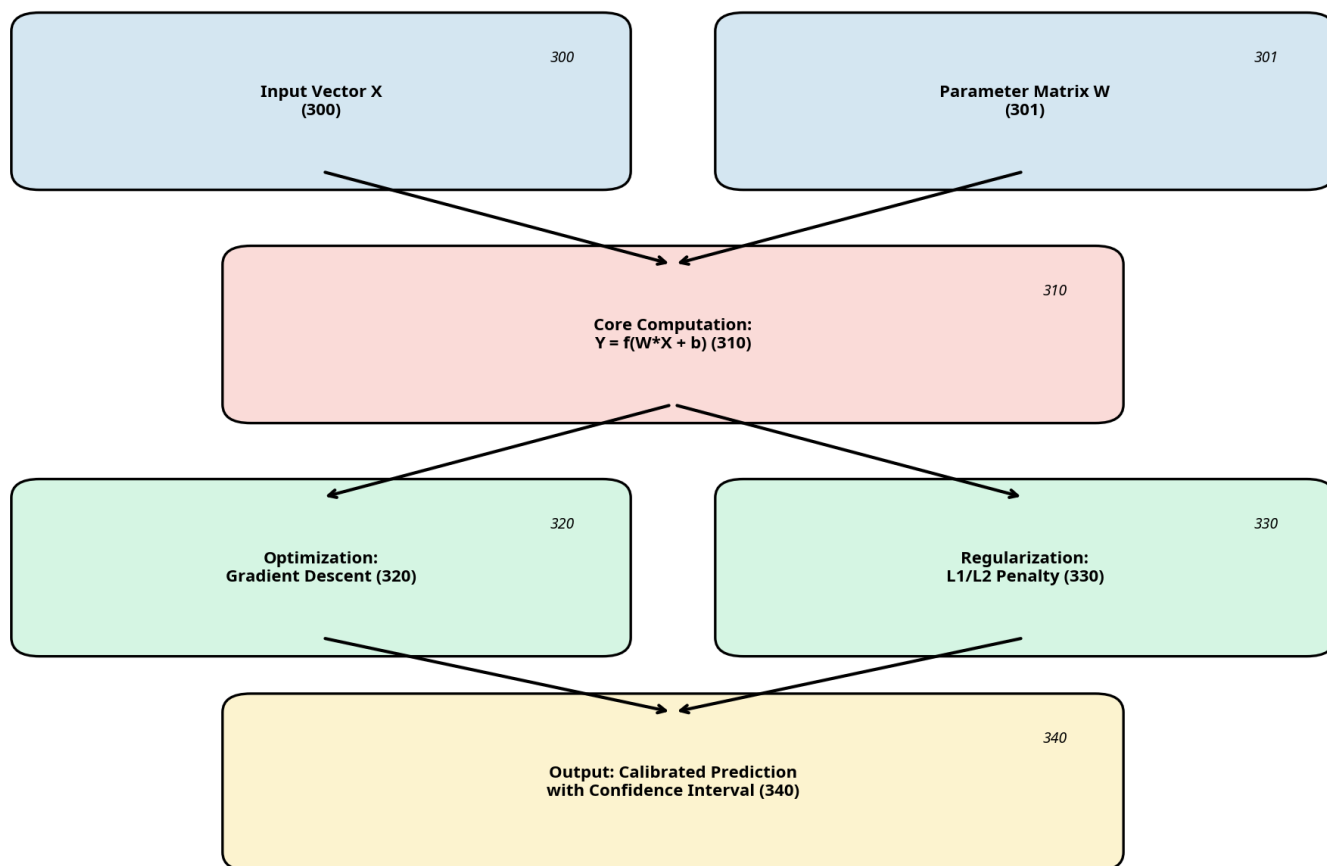


FIG. 3 - Vital Signs Monitor Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - Vital Signs Monitor Performance Validation

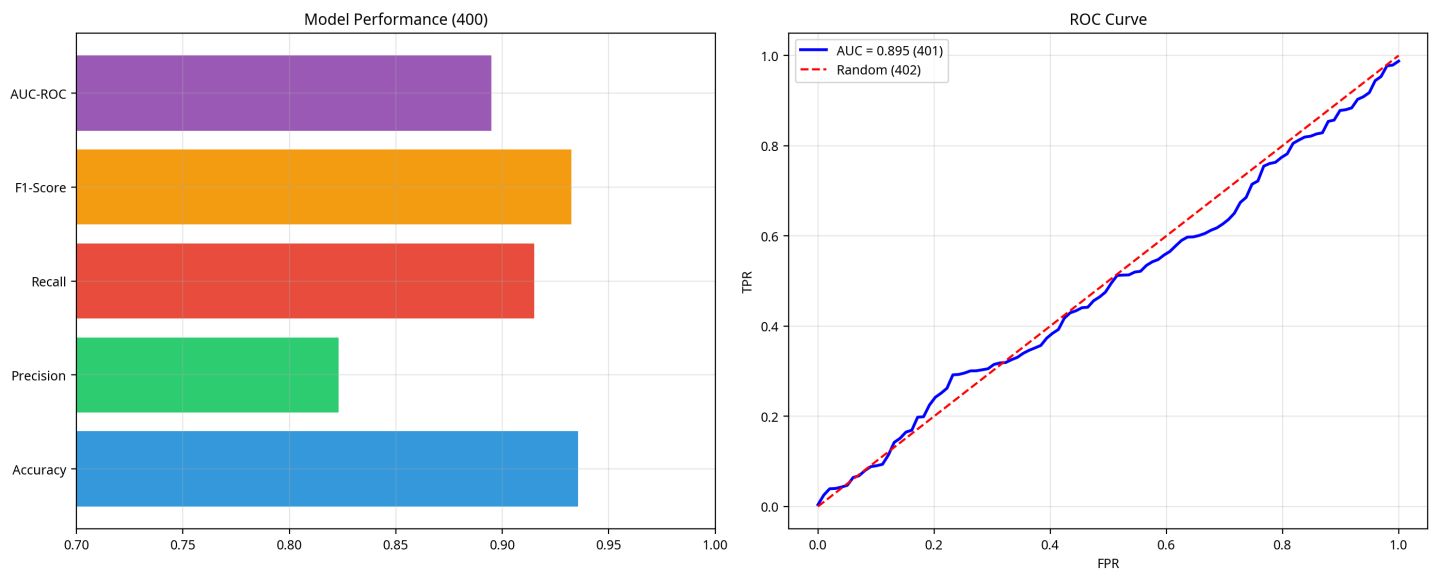


FIG. 5 - Vital Signs Monitor vs. Prior Art Comparison

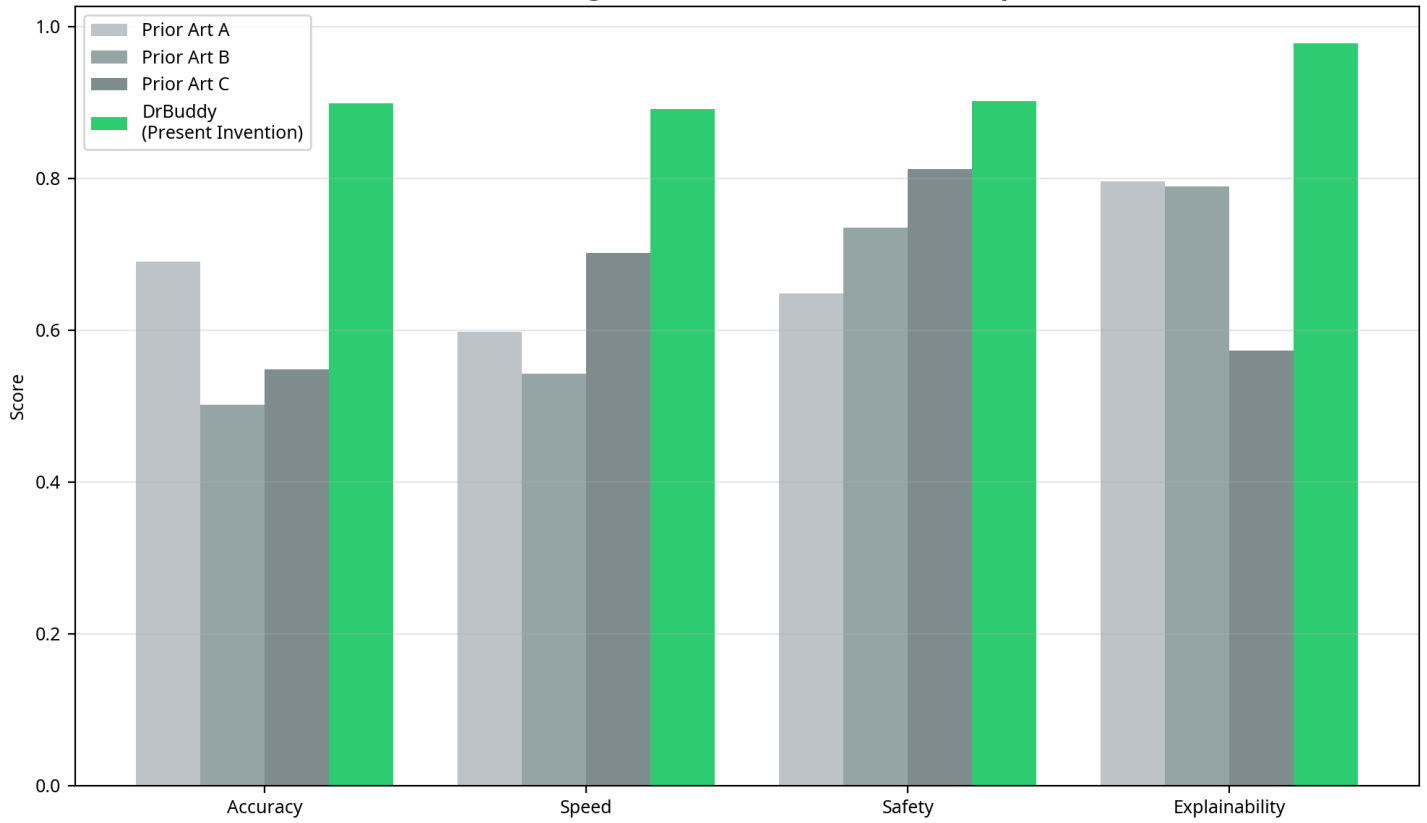


FIG. 6 - Vital Signs Monitor Safety Verification Pipeline

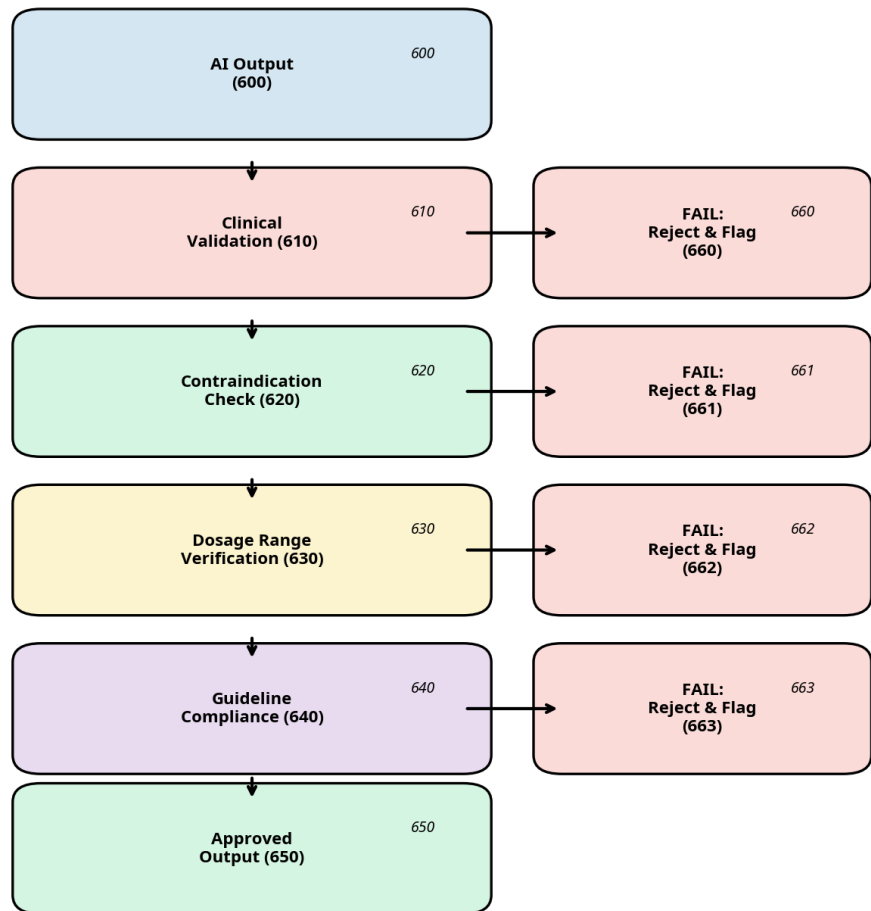


FIG. 7 - Vital Signs Monitor Hardware & Cloud Infrastructure

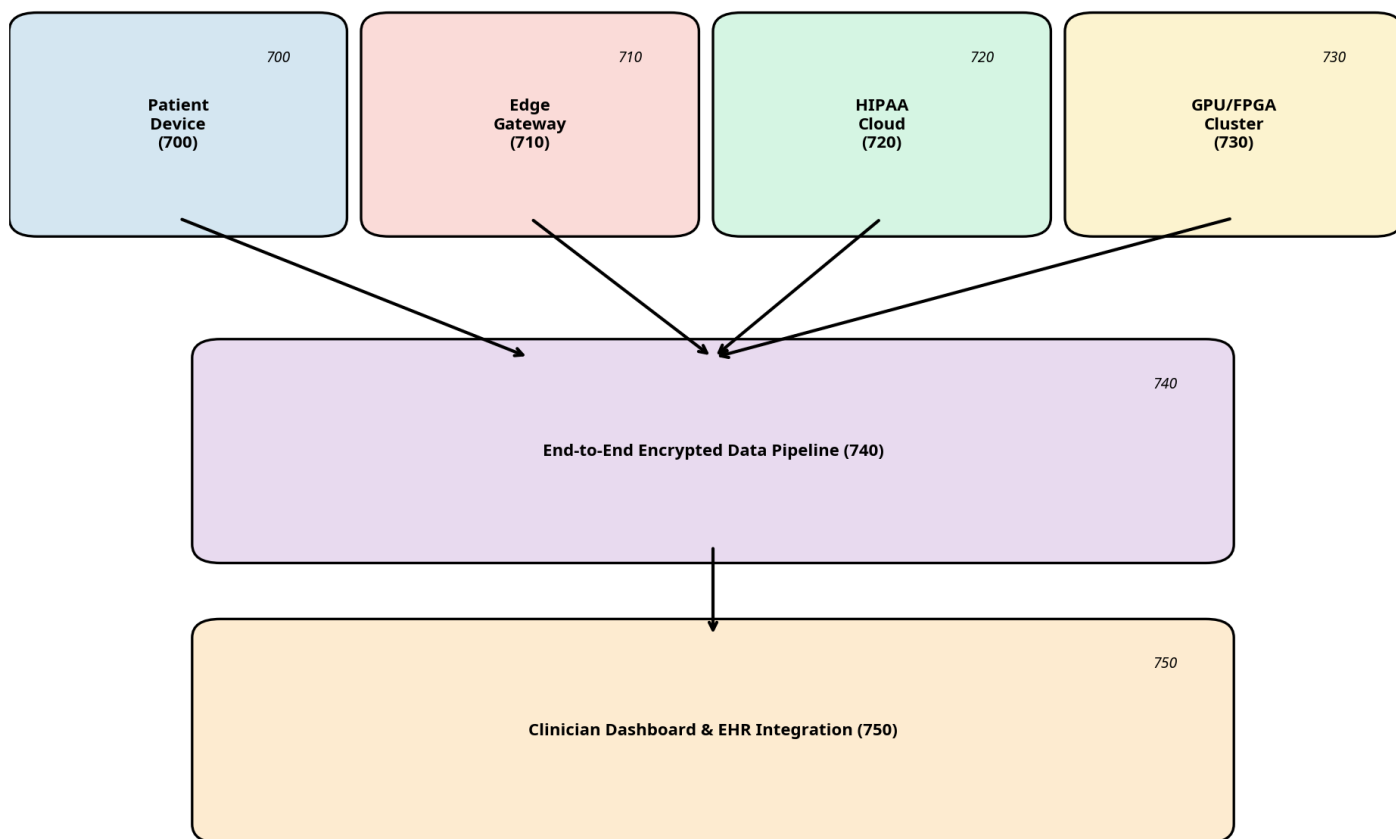


FIG. 8 - Vital Signs Monitor Scalability Benchmark

